

Crowbar Evaltree Insights

AI Safety — 2025 Edition

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Executive Summary

AI safety in 2025 refers to technical and organizational practices designed to reduce risks from artificial intelligence systems. These risks range from model outputs that contain harmful content or misinformation, to broader concerns about autonomous systems acting in ways misaligned with human intent.

The field has moved from theoretical discussion to regulatory action. The European Union's AI Act entered into force in August 2024 and began phased implementation. The United States issued its Executive Order on Safe, Secure, and Trustworthy Artificial Intelligence in October 2023, establishing reporting requirements for frontier AI developers. China, the United Kingdom, India, and other jurisdictions have introduced guidelines, standards, or binding rules.

Key themes in 2025 include the operationalization of risk-based regulatory frameworks, increased transparency requirements for foundation model developers, growing attention to dual-use risks from publicly available models, and divergence between jurisdictions on how to classify and govern AI systems.

This brief synthesizes public information on what AI safety means, what risks regulators and researchers identify, what rules currently exist, and what organizations are doing in response. It is not advice and takes no position on policy outcomes.

What Is AI Safety (Plain English)

AI safety is the practice of building, deploying, and governing AI systems to reduce the probability and severity of harmful outcomes. The term encompasses technical research, engineering practices, organizational governance, and regulatory compliance.

Scope

AI safety includes model alignment (ensuring AI systems behave as intended), robustness (resistance to adversarial inputs or distributional shift), interpretability (understanding why models produce specific outputs), and misuse prevention (reducing the risk that capable systems are used for harmful purposes).

It also includes concerns about systemic risks such as economic disruption, the concentration of power, and long-term risks from highly capable autonomous systems. Different stakeholders emphasize different aspects. Regulators focus on near-term harms and accountability. Researchers emphasize technical challenges in aligning advanced models. Industry actors prioritize operational risk management and reputation.

Why the term is often misunderstood

AI safety is sometimes conflated with AI ethics, but the two are distinct. Ethics addresses values, fairness, and societal impact. Safety focuses on preventing specific harmful behaviors or failure modes. AI safety is also distinct from cybersecurity, though the two overlap when considering adversarial attacks on models.

The term has been used to describe everything from content moderation on chatbots to existential risks from superintelligent systems. This breadth creates confusion. Some critics argue the term is too vague to be useful. Supporters argue it captures a necessary interdisciplinary field.

The AI Safety Institute established by the UK government in November 2023 defines its mission as evaluating risks from advanced AI systems and supporting the development of safety standards (UK AI Safety Institute, <https://www.aisi.gov.uk>). The U.S. AI Safety Institute, announced in the same period and established under the National Institute of Standards and Technology, focuses on developing measurement science and guidelines for trustworthy AI (NIST AI Safety Institute Consortium, <https://www.nist.gov/aisi>).

Core AI Safety Risk Categories

Public sources identify four broad risk categories. These are not exhaustive but reflect consensus among regulators and research institutions.

Model Behavior & Alignment

Advanced AI models may produce outputs that are harmful, deceptive, or inconsistent with developer intent. This includes generating illegal content, providing instructions for dangerous activities, or exhibiting emergent capabilities not anticipated during training.

The U.S. National Institute of Standards and Technology's AI Risk Management Framework identifies risks from "harmful bias and homogenization" and "dangerous or violent recommendations" (NIST AI RMF 1.0, January 2023, <https://www.nist.gov/it1/ai-risk-management-framework>). Anthropic's Responsible Scaling Policy, published in September 2023, describes capability thresholds at which models may pose risks such as enabling weapons development or autonomous self-replication (Anthropic RSP, <https://www.anthropic.com/news/responsible-scaling-policy>).

Alignment research seeks to ensure models reliably pursue intended goals. Techniques include reinforcement learning from human feedback, constitutional AI, and interpretability research. These remain active research areas with no settled solutions.

Data Risks

Training data may contain copyrighted material, personal information, or biased patterns that models reproduce. Models may memorize and regurgitate training data, creating privacy or intellectual property risks.

The European Union's General Data Protection Regulation applies to AI systems that process personal data. The EU AI Act, which entered into force on August 1, 2024, requires transparency about training data for high-risk systems (EU AI Act, Regulation (EU) 2024/1689, <https://eur-lex.europa.eu>). The U.S. Federal Trade Commission has signaled enforcement interest in AI systems that violate consumer protection laws through biased or deceptive outputs (FTC AI Blog, <https://www.ftc.gov/business-guidance/blog/topics/artificial-intelligence>).

Data risks also include dataset poisoning, where adversaries introduce malicious data to corrupt model behavior, and model inversion attacks, where attackers extract training data from deployed models.

Deployment & Misuse Risks

Capable AI systems may be misused for spam, disinformation, cyberattacks, or the creation of chemical, biological, radiological, or nuclear weapons. Open-source models present a governance challenge because they cannot be recalled or access-controlled after release.

The U.S. Executive Order 14110, issued in October 2023, requires developers of foundation models trained with significant computational resources to report safety test results to the Department of Commerce (White House EO 14110, <https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/>). The order specifically addresses dual-use foundation models that could enable weapons of mass destruction.

The RAND Corporation published research in 2023 on how large language models might assist non-experts in planning biological attacks, concluding that current models provide marginal assistance but future models may lower barriers (RAND, "Operational Risks of AI in Large-Scale Biological Attacks", https://www.rand.org/pubs/research_reports/RRA2977-2.html).

Systemic and Societal Risks

AI systems may create risks at the system level that are not visible at the individual model level. These include labor market disruption, the spread of misinformation at scale, concentration of market power among a small number of model developers, and the potential for AI systems to destabilize critical infrastructure or financial systems.

The OECD AI Principles, adopted in 2019 and updated in 2023, call for AI systems that are robust, secure, and safe throughout their lifecycle, with particular attention to societal impacts (OECD AI Principles, <https://oecd.ai/en/ai-principles>). The United Nations established an Advisory Body on AI in 2023, which issued interim recommendations in December 2023 calling for international cooperation on AI governance (UN Secretary-General's AI Advisory Body, <https://www.un.org/en/ai-advisory-body>).

Long-term risks from highly autonomous systems remain speculative but are taken seriously by some research institutions. The Centre for the Governance of AI and the Future of Humanity Institute have published research on tail risks, though many of these scenarios lack empirical grounding.

Regulatory & Policy Landscape

Regulatory approaches vary significantly by jurisdiction. The following reflects the state of binding rules and published guidance as of January 2025.

European Union

The EU AI Act is the most comprehensive binding AI regulation in force. It entered into force on August

1, 2024, with phased implementation over 24 to 36 months. The Act takes a risk-based approach, categorizing AI

systems into four levels: unacceptable risk (banned), high risk (subject to conformity assessments and transparency requirements), limited risk (subject to transparency obligations), and minimal risk (no specific requirements).

Prohibited practices include social scoring by governments, real-time biometric identification in public spaces (with narrow exceptions), and manipulative or exploitative AI systems. High-risk systems include those used in critical infrastructure, education, employment, law enforcement, and border control.

General-purpose AI models with systemic risk (those trained with computational power exceeding 10^{25} FLOPs) face additional obligations, including adversarial testing, systemic risk evaluation, and incident reporting. Developers must publish summaries of training data (EU AI Act, Articles 52-55, <https://eur-lex.europa.eu>).

The European AI Office, established within the European Commission, is responsible for enforcement and coordination. Member states are designating national competent authorities.

United States

The U.S. lacks comprehensive federal AI legislation but has introduced binding requirements through executive action and agency rulemaking.

Executive Order 14110, issued in October 2023, requires developers of foundation models trained using more than 10^{26} integer or floating-point operations to share safety test results and other information with the Department of Commerce. The order directs NIST to develop standards and the Department of Homeland Security to address AI risks in critical infrastructure (White House EO 14110, <https://www.whitehouse.gov>).

The Office of Management and Budget issued guidance in March 2024 requiring federal agencies to assess and manage risks from AI systems that impact rights or safety (OMB M-24-10, <https://www.whitehouse.gov/omb/briefing-room/2024/03/28/omb-releases-implementation-guidance-following-president-bidens-executive-order-on-artificial-intelligence/>).

Several states have introduced or passed AI-related legislation. California's AI Transparency Act, effective January 2024, requires disclosures for AI-generated content in political advertising. Colorado enacted legislation in 2024 requiring impact assessments for high-risk AI systems (Colorado SB 24-205, <https://leg.colorado.gov>).

The U.S. Congress has introduced multiple AI bills but none have passed as of January 2025. Proposals include algorithmic accountability requirements, liability frameworks, and funding for AI safety research.

United Kingdom

The UK has adopted a principles-based, pro-innovation approach to AI regulation. The government published a white paper in March 2023 outlining five principles: safety, security, and robustness;

transparency and explainability; fairness; accountability and governance; and contestability and redress (UK AI White Paper,

<https://www.gov.uk/government/publications/ai-regulation-a-pro-innovation-approach>).

Rather than creating a new regulator, the UK assigns responsibility to existing regulators such as the Information Commissioner's Office, the Competition and Markets Authority, and sector-specific bodies. These regulators issued guidance in 2024 on applying existing laws to AI systems.

The UK AI Safety Institute, announced at the AI Safety Summit in November 2023, focuses on evaluating advanced AI models and publishing safety research. It has signed memoranda of understanding with leading AI developers to provide pre-deployment access to models for safety testing (UK AI Safety Institute, <https://www.aisi.gov.uk>).

India

India's approach to AI governance emphasizes innovation while addressing specific harms. The Ministry of Electronics and Information Technology issued an advisory in March 2024 requiring platforms deploying AI models to obtain government approval before public release if the models are unreliable or undertested. The advisory was revised in March 2024 to clarify that approval applies only to models not yet deployed (MeitY Advisory, <https://www.meity.gov.in>).

India's Digital Personal Data Protection Act, enacted in 2023, applies to AI systems that process personal data. The law includes provisions for data minimization, purpose limitation, and user rights.

India has participated in international AI governance discussions, including at the AI Safety Summit and through bilateral agreements with the United States and the European Union on AI cooperation.

Other Major Jurisdictions

China has implemented several AI-specific regulations, including the Algorithm Recommendation Management Provisions (2022), which require algorithmic transparency and user opt-out options, and the Generative AI Measures (2023), which impose content moderation and data security requirements on generative AI services (CAC Generative AI Measures, <http://www.cac.gov.cn>).

Canada introduced the Artificial Intelligence and Data Act as part of Bill C-27 in 2022. The bill establishes requirements for high-impact AI systems, including risk assessments, record-keeping, and incident reporting. It has not yet been enacted (Canadian AIDA, <https://www.parl.ca/DocumentViewer/en/44-1/bill/C-27/first-reading>).

Singapore has published a Model AI Governance Framework and an Implementation and Self-Assessment Guide for Organizations, both voluntary. The framework emphasizes explainability, transparency, and human oversight (Singapore Model AI Governance Framework, <https://www.pdpc.gov.sg>).

Brazil is developing AI legislation. A bill approved by the Senate in 2024 would establish a risk-based framework similar to the EU AI Act, with additional provisions for liability and transparency (Brazilian AI Bill PL 2338/2023, <https://www.camara.leg.br>).

Industry & Standards Responses

Private sector actors have developed voluntary frameworks, industry standards, and self-regulatory commitments in response to regulatory pressure and reputational concerns.

Safety Frameworks

Several AI developers have published safety frameworks describing their approach to risk management.

OpenAI's Preparedness Framework, published in December 2023, defines risk categories (cybersecurity, chemical/biological/radiological/nuclear, persuasion, model autonomy) and risk levels (low, medium, high, critical). The framework requires models to remain below high risk in all categories before deployment (OpenAI Preparedness Framework, <https://openai.com/safety/preparedness>).

Anthropic's Responsible Scaling Policy, updated in September 2023, sets capability thresholds (AI Safety Levels) at which models may pose catastrophic risks. The policy commits to implementing safeguards proportional to assessed risk, including security measures, alignment techniques, and deployment restrictions (Anthropic RSP, <https://www.anthropic.com/news/responsible-scaling-policy>).

Google DeepMind published its Frontier Safety Framework in May 2024, describing a risk-based approach with critical capability evaluations and defined risk thresholds. The framework commits to not deploying models that exceed risk thresholds without adequate mitigations (Google DeepMind Frontier Safety Framework, <https://deepmind.google/discover/blog/introducing-the-frontier-safety-framework/>).

These frameworks are voluntary commitments. They are not legally binding and may be revised.

Auditing

Third-party auditing of AI systems is an emerging practice. Several organizations offer AI audit services, though standards for what constitutes an audit vary.

The UK AI Safety Institute and the U.S. AI Safety Institute have conducted evaluations of frontier models, though detailed results are typically not published. The institutes aim to develop standardized evaluation protocols.

The EU AI Act requires conformity assessments for high-risk AI systems, conducted by notified bodies accredited by member states. The Commission is developing technical standards to support these assessments.

Industry groups such as the Partnership on AI and the Frontier Model Forum have worked on shared evaluation benchmarks, though adoption remains uneven (Frontier Model Forum, <https://www.frontiermodelforum.org>).

Model Governance

Model governance refers to organizational practices for managing AI development and deployment risks.

The NIST AI Risk Management Framework provides a voluntary structure for identifying, assessing, and managing AI risks. It includes processes for mapping risks to context, measuring risk levels, and implementing

controls (NIST AI RMF, <https://www.nist.gov/itl/ai-risk-management-framework>).

The International Organization for Standardization and the International Electrotechnical Commission published ISO/IEC 42001 in December 2023, a management system standard for AI. The standard provides requirements for establishing, implementing, and improving an AI management system (ISO/IEC 42001, <https://www.iso.org/standard/81230.html>).

Model cards and system cards are used by some developers to document model capabilities, limitations, and intended use. These documents are intended to inform downstream users but vary widely in detail and format.

Industry Self-Regulation Efforts

In July 2023, major AI developers including OpenAI, Google, Microsoft, and Anthropic made voluntary commitments to the White House on AI safety. These included red-teaming models before release, sharing information on managing AI risks, investing in cybersecurity, and developing tools to detect AI-generated content (White House Voluntary AI Commitments, <https://www.whitehouse.gov/briefing-room/statements-releases/2023/07/21/fact-sheet-biden-harris-administration-secures-voluntary-commitments-from-leading-artificial-intelligence-companies-to-manage-the-risks-posed-by-ai/>).

The Frontier Model Forum, launched in July 2023 by OpenAI, Google DeepMind, Anthropic, and Microsoft, is an industry body focused on advancing AI safety research and promoting responsible development. The forum has announced an AI Safety Fund and published technical reports on evaluations (Frontier Model Forum, <https://www.frontiermodelforum.org>).

These efforts are voluntary and not independently verified.

What Organizations Are Actually Doing

Public disclosures indicate the following practices among frontier AI developers.

Red-teaming: Most major developers conduct internal adversarial testing before model release. Some engage external researchers or host public bug bounty programs. OpenAI, Google, Anthropic, and Meta have all described red-teaming efforts in public documentation.

Access controls: Developers of closed-source models implement usage policies and rate limits to reduce misuse risk. OpenAI, Anthropic, and Google restrict access to certain capabilities or deny requests that violate acceptable use policies. Enforcement is primarily automated with human review for edge cases.

Capability evaluations: Developers assess models for dangerous capabilities such as assisting with cyberattacks, producing hazardous biological or chemical information, or exhibiting deceptive behavior. Evaluations vary in rigor. Some are conducted internally; others involve collaboration with external researchers or government institutes.

Post-deployment monitoring: Several developers monitor deployed models for misuse or unexpected behavior. Meta's Llama models, released under permissive licenses, have limited post-deployment oversight because the models can be run locally.

Transparency reports: Some developers publish transparency reports describing use-policy enforcement actions. OpenAI and Anthropic have published limited data on policy violations and account suspensions.

Collaboration with regulators: Developers have shared information with the U.S. AI Safety Institute and the UK AI Safety Institute under voluntary agreements. The extent and detail of these collaborations are not fully public.

Safety research: Companies fund internal safety research teams and support external research. DeepMind, OpenAI, and Anthropic have published research on alignment, interpretability, and adversarial robustness. Academic institutions such as UC Berkeley's Center for Human-Compatible AI and the Machine Intelligence Research Institute also conduct safety research.

What is not being done, or not disclosed:

Independent verification of safety claims is rare. Third-party audits are not standard practice. Many details of safety measures are considered proprietary. Incident reporting requirements exist under some regulations but are not yet routine. There is no public registry of AI safety incidents comparable to aviation or pharmaceutical adverse event databases.

Common Misconceptions About AI Safety

Misconception 1: AI safety is only about existential risk.

AI safety encompasses near-term harms such as bias in hiring algorithms, misinformation at scale, and privacy violations. These risks affect people today. Long-term risks from highly autonomous systems are a subset of AI safety, not the whole field.

Misconception 2: AI safety means slowing down AI development.

Safety measures can be implemented without halting progress. Many safety techniques, such as improved training methods and better evaluation, increase model quality. Regulatory frameworks like the EU AI Act permit innovation while requiring risk management for high-risk systems.

Misconception 3: Open-source models are inherently less safe.

Open-source models offer transparency and allow independent scrutiny, which can improve safety. However, they cannot be recalled or access-controlled after release, creating governance challenges for dual-use risks. The safety of a model depends on design, testing, and deployment context, not solely on whether it is open or closed.

Misconception 4: AI safety is solved by existing cybersecurity or software engineering practices.

AI systems introduce novel failure modes not present in traditional software. Model outputs are probabilistic and context-dependent. Adversarial attacks exploit statistical properties of neural networks. Safety practices must address these unique characteristics.

Misconception 5: Regulation will eliminate AI risk.

Regulation sets minimum standards and allocates accountability, but it cannot prevent all harms. Technical research, organizational governance, and cultural norms all play roles. Over-reliance on compliance can create a false sense of security.

Misconception 6: AI safety is a primarily technical problem.

AI safety requires technical solutions, but also organizational design, legal frameworks, and societal choices about acceptable risk. Purely technical approaches miss governance and accountability challenges.

What to Watch Going Forward

The following developments are likely in the near and medium term based on published regulatory timelines and publicly stated organizational priorities.

Near-term (2025-2026)

The EU AI Act will move into full implementation. Notified bodies will begin conformity assessments. Enforcement actions are likely by late 2025 or early 2026. Developers serving EU markets will need to demonstrate compliance.

The U.S. AI Safety Institute will publish evaluation standards and guidelines. Voluntary participation is expected, but federal procurement requirements may create de facto compliance pressure.

Several jurisdictions, including Canada and Brazil, may enact new AI legislation. India and Southeast Asian countries are expected to clarify regulatory requirements.

Liability questions will become more prominent as AI systems are used in high-stakes contexts. Courts in multiple jurisdictions will adjudicate cases involving algorithmic harm, setting precedents for accountability.

Industry standards bodies, including ISO and IEEE, will publish additional AI standards. Adoption will be voluntary but may be referenced in regulations or procurement requirements.

Medium-term (2026-2028)

International coordination on AI safety may increase. The OECD, UN, and regional organizations are expected to develop norms and potentially binding agreements on issues such as dual-use model

governance and cross- border data flows.

Capability evaluations are likely to become more standardized and rigorous. Benchmark datasets and evaluation protocols may converge, enabling comparisons across models and organizations.

Post-deployment monitoring and incident reporting may become routine, especially for high-risk systems. Industry or regulatory bodies may establish incident databases.

The effectiveness of current safety measures will become clearer as more capable models are deployed. Failures or high-profile incidents may prompt regulatory or industry responses.

Uncertainty

The pace of AI capability improvement is uncertain. If capabilities plateau, safety concerns may shift from preventing misuse of highly capable systems to managing widespread deployment of moderately capable systems. If capabilities accelerate, dual-use risks and alignment challenges may intensify.

The balance between open and closed models is unresolved. Policy choices on model openness will shape the distribution of AI capabilities and the feasibility of governance.

International regulatory divergence could create compliance complexity for global developers or fragment the AI ecosystem. Alternatively, convergence toward shared standards could emerge.

Public perception of AI risk may shift based on real-world incidents or media coverage. Perception influences political will for regulation and organizational investment in safety.

Conclusion

AI safety in 2025 is characterized by regulatory implementation, industry self-regulation, and ongoing technical research. The field spans near-term harms such as bias and misinformation, dual-use risks from capable models, and longer-term concerns about alignment and autonomy.

Regulators in the European Union, United States, United Kingdom, and other jurisdictions have introduced binding rules or guidance. Industry actors have published voluntary commitments and safety frameworks. Research institutions continue to investigate technical and governance challenges.

Significant uncertainty remains about the effectiveness of current measures, the trajectory of AI capabilities, and the coordination of international governance. Organizations deploying AI systems face complex compliance requirements and evolving best practices.

This brief synthesizes publicly available information as of January 2025. It takes no position on policy outcomes and offers no advice. Readers should consult legal, technical, and regulatory professionals for decisions specific to their context.

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End of Intelligence Brief

This document was prepared using publicly available sources as of January 2025. Information is subject to change as regulations evolve and new research emerges.